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Definition 1: Let (X, Σ, μ) be a measure space. Let $\{f_n\}_{n\in\mathbb{N}}$ be a sequence of real measurable functions, and $M \in \Sigma$ such that $\mu(M) = 0$. If $f_n(x) \to f(x)$ for all $x \in M^c$, for some measurable function f then $\{f_n\}_{n\in\mathbb{N}}$ is said to converge to f **almost everywhere**.

Definition 2: Let (X, Σ, μ) be a measure space. Let $\{f_n\}_{n\in\mathbb{N}}$ be a sequence of real measurable functions,then $\{f_n\}_{n\in\mathbb{N}}$ is said to converge to f **in measure** if, for every $\epsilon > 0$

$$\lim_{n\to\infty} \mu(\{x \in X : |f_n(x) - f(x)| \geq \epsilon\}) = 0$$

Theorem 1: Let (X, Σ, μ) be a finite measure space. Let $\{f_n\}_{n\in\mathbb{N}}$ be a sequence of real measurable functions. If f_n converges to f almost everywhere, then f_n converges to f in measure.

Proof: Let f_n converges to f almost everywhere. Consider $\epsilon > 0$, define $\{A_n\}_{n\in\mathbb{N}}$ as

$$A_n = \{x \in X : |f_n(x) - f(x)| \geq \epsilon\}$$

Notice, if $x \in M^c$ then $\exists\ N_x \in \mathbb{N}$ such that $x \notin A_j\ \forall j \geq N_x$, i.e. $x \notin \bigcup\limits_{j\geq N_x} A_j$.

$$\implies \bigcap_{n=1}^{\infty} \bigcup_{j\geq n} A_j \subseteq M$$

By Countable Additivity and Finiteness of μ we conclude,

$$0 \leq \lim_{n\to\infty} \mu(A_n) \leq \lim_{n\to\infty} \mu(\cup_{j\geq n} A_j) = \mu(\cap_{n\in\mathbb{N}} \cup_{j\geq n} A_j) \leq \mu(M) = 0$$

$$\implies \lim_{n\to\infty} \mu(A_n) = 0$$

Since $\epsilon > 0$ is arbitrary we conclude $\{f_n\}_{n\in\mathbb{N}}$ converges to f in measure.

Theorem 2: Let (X, Σ, μ) be a finite measure space. Let $\{f_n\}_{n\in\mathbb{N}}$ be a sequence of real measurable functions. If f_n converges to f in measure, then there is a subsequence of f_n that converges to f almost everywhere.

Proof: Let f_n converges to f in measure. Let $\epsilon_n = \frac{1}{n}$, Since f_n converges to f almost everywhere we construct $\{n_j\}_{j\in\mathbb{N}} \subseteq \mathbb{N}$ such that,

1. $n_i < n_{i+1}\ \forall i \in \mathbb{N}$
2. $\mu(|f_{n_i} - f| \geq \epsilon_i) < \frac{1}{2^i}$

Let $A_i = \{x \in X : |f_{n_i}(x) - f(x)| \geq \epsilon_i\}$ and $M = \bigcap_{n=1}^{\infty} \bigcup_{j\geq n} A_j$. Clearly,

$$\mu(M) = \lim_{n\to\infty} \mu(\cup_{j\geq n} A_j) \leq \lim_{n\to\infty} \sum_{j=n}^{\infty} \mu(A_j) = \lim_{n\to\infty} \sum_{j=n}^{\infty} \frac{1}{2^j}$$

As $\{2^{-j}\}_{j\in\mathbb{N}}$ is a convergent series, we know the limit on RHS is just the trailing limit and is equal to zero so, $\mu(M) = 0$. Now, consider the sequence $\{f_{n_j}\}_{j\in\mathbb{N}}$, let $x \in M^c$

$$\implies x \in \bigcup_{n=1}^{\infty} \bigcap_{j\geq n} A_j^c$$

$$\implies x \in \bigcap_{j\geq N_1} A_j^c \text{ for some } N_1 \in \mathbb{N}$$

$$\implies \forall j \geq N_1\ |f_{n_j}(x) - f(x)| < \frac{1}{j}$$

Fix $\epsilon > 0$, by Archemidean property we know $\exists N_2 \in \mathbb{N}$ such that $\frac{1}{N_2} < \epsilon$. Fix $N = \max(N_1, N_2)$ then for all $j \geq N$

$$|f_{n_j}(x) - f(x)| < \frac{1}{j} \leq \frac{1}{N} < \epsilon$$

Since $\epsilon > 0$ and $x \in M^c$ are arbitrary, we conclude.

$$\lim_{j\to\infty} f_{n_j}(x) = f(x) \quad \forall x \in M^c$$

and $\mu(M) = 0$ So, $\{f_{n_j}\}_{j\in\mathbb{N}}$ is the subsequence that converges to f almost everywhere.